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DESIGN AN IMPLEMENTATION OF IRS FOR SMALL MIMO WIRELESS NETWORK

A PROJECT REPORT

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ABSTRACT

The **exponential growth of wireless data demand** and the **limitations of conventional communication infrastructure** have necessitated the development of **innovative technologies** to improve **signal quality** and **energy efficiency**. **Intelligent Reflecting Surfaces (IRS)** have emerged as a **promising solution** in **next-generation wireless networks**.

This project explores the **design and implementation** of an **IRS-assisted wireless communication system**, where a **programmable meta surface** is used to **dynamically control** the reflection of incident signals toward the receiver, thereby **enhancing the overall communication performance**.

The **IRS is modelled using discrete phase shifts** and is integrated into a **simulation environment** to analyse its impact on system parameters such as **signal-to-noise ratio (SNR)**, **bit error rate (BER)**, and **achievable data rate**. **Optimization algorithms** are employed to configure the IRS elements in real-time, **maximizing signal strength** at the receiver while **minimizing interference**.

The project demonstrates that **IRS can significantly improve wireless coverage, reliability, and spectral efficiency**, especially in **non-line-of-sight (NLoS) scenarios**.

This work contributes toward the **practical realization of IRS in 6G networks**, highlighting its potential in **future smart radio environments**.

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	iv
	LIST OF TABLES	viii
	LIST OF FIGURES	ix
	LIST OF ABBREVIATIONS	x
1	INTRODUCTION	1
	1.1 OVERVIEW OF THE PROJECT	1
	1.2 WIRELESS COMMUNICATION	3
	1.3 WIRELESS TECHNOLOGY	3
	1.4 WIRELESS COMMUNICATION STANDARDS	4
	1.5 5G / 6G GENERATION	5
	1.6 MIMO AND ITS TYPES	6
	1.7 IRS DETAILS	7
	1.8 COMSOL MULTIPHYSICS 6.3	8
	1.9 SCOPE OF THE PROJECT	9
	1.10 OBJECTIVE	10
2	LITERATURE SURVEY	11
3	EXISTING AND PROPOSED SYSTEM	15
	3.1 EXISTING SYSTEM	15
	3.1.1 DISADVANTAGES	15
	3.2 PROPOSED SYSTEM	17
	3.2.1 ADVANTAGES	17
4	SYSTEM DESIGN	19
	4.1 BLOCK DIAGRAM	19
	4.2 DESCRIPTION	20

5	HARDWARE REQUIREMENTS	23
	5.1 CAPACITOR	24
	5.2 INDUCTOR	24
	5.3 VARACTOR DIODE	25
	5.4 IRS PANEL	25
	5.5 RF TRANSMITTER & RECEIVER	27
	5.5.1 RF TRANSMITTER	27
	5.5.2 RF RECEIVER	28
	5.6 TOWER	29
	5.7 ANTENNA	30
	5.8 FPGA	32
6	SOFTWARE REQUIREMENTS	34
	6.1 COMSOL MULTIPHYSICS	34
	6.1.1 SYSTEM COMPONENT	34
	6.1.2 APPLICATIONS	36
	6.1.3 FUNCTIONS	38
	6.2 MATLAB	40
	6.2.1 FUNCTIONS	40
	6.2.2 WHY MATLAB?	40
	6.2.3 MATLAB APPLICATIONS	41
7	SYSTEM IMPLEMENTATION	42
	7.1 SYSTEM IMPLEMENTATION	42
	7.2 KEY POINTS OF SYSTEM IMPLEMENTATION	44
8	EXPERIMENTAL RESULT AND ANALYSIS	45
	8.1 DESIGN OF IRS USING COMSOL MULTIPHYSICS SOFTWARE	46
	8.2 MATLAB SIMULATION AND RESULTS	48
9	CONCLUSION	51
10	FUTURE SCOPE	53

APPENDIX	54
REFERENCES	66
FIELD VISIT CERTIFICATE	69
CONFERENCE CERTIFICATE	70

LIST OF TABLES

TABLE NO.	TITLE	PAGE NO.
5.8.1	FPGA SPECIFICATION	33

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
1.4	EVALUATION OF 1G TO 6G WIRELESS COMMUNICATION	5
4.1	BLOCK DIAGRAM	19
5.1	CAPACITOR	24
5.2	INDUCTOR	24
5.3	VARACTOR DIODE	25
5.4	IRS PANEL	25
5.5	RF TRANSMITTER AND RECEIVER	27
5.6	COMMUNICATION TOWER	29
5.7	ANTENNA	30
7.1.1	MATLAB CODE FOR IRS	42
7.1.2	POLAR PLOTS OF PHASE SHIFTS AND SNR RATIO	43
8.1.1	IRS ARRAY	46
8.1.2	SINGLE ARRAY OUTLINE	47
8.1.3	IRS PANEL DESIGN	47
8.1.4	RADIATION PATTERN OF IRS PANEL	48
8.2.1	PERFORMANCE: WITH IRS AND WITHOUT IRS	49
8.2.2	IRS PHASE SHIFTS	49
8.2.3	SNR COMPARISON: WITH IRS VS WITHOUT IRS	50
8.2.4	POWER CONTRIBUTIONS ANALYSIS	50

ABBREVIATIONS

S. NO.	ABBREVIATION	EXPANSION
1	IRS	Intelligent Reflecting Surface
2	MIMO	Multiple Input Multiple Output
3	IoT	Internet Of Things
4	NLOS	Non-Line-Of-Sight
5	SNR	Signal-to-Noise Ratio
6	FPGA	Field-Programmable Gate Array
7	RF	Radio Frequency
8	WSN	Wireless Sensor Network
9	CSI	Channel State Information
10	AI	Artificial Intelligence
11	ML	Machine Learning
12	WPCN	Wireless Powered Communication Network
13	ISWPT	Integrated Sensing and Wireless Power Transfer
14	SDP	Semi-Definite Programming
15	LC	Low Complexity
16	MM	Majorization-Minimization
17	HAP	High-Altitude Platform
18	WD	Wireless Devices
19	AR/VR	Augmented Reality / Virtual Reality
20	FEA	Finite Element Analysis
21	GUI	Graphical User Interface
22	EMC	Electro-Magnetic Compatibility
23	MEMS	Micro-Electro-Mechanical Systems
24	SISO	Single Input Single Output
25	OFDM	Orthogonal Frequency Division Multiplexing
26	URLLC	Ultra-Reliable Low-Latency Communication

27	Wi-Fi	Wireless Fidelity
28	AM	Amplitude Modulation
29	FM	Frequency Modulation
30	PSK	Phase Shift Keying
31	FSK	Frequency Shift Keying
32	APES	Amplitude and Phase Estimation